Calibrated Uncertainty and Explicit Costs for Action Selection in Tabular Data Cleaning

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Abstract*—Automated data repair systems are normally judged on how accurately they correct values. We argue, and show empirically, that the harder and more consequential problem is deciding whether to act at all. We present a pipeline that attaches two calibrated probabilities to every flagged cell — the probability that a proposed repair is correct, and the probability that the cell is genuinely corrupted — and selects among three actions (automatic repair, human review, abstention) by minimising expected cost. On three benchmark datasets the resulting policy is the only one of five candidate policies that is never badly beaten, attaining zero worst-case regret, and this holds when every clean-data advantage is removed from the detector. We then report a negative result we consider equally important: five successive mechanisms for identifying which column is corrupted all failed, each absorbing a different column-specific nuisance quantity rather than the corruption signal. Using a Bayes-optimal ceiling computed from a known corruption process, we show that within a corrupted column our detector already captures 77.9% of the attainable signal, so the residual loss is almost entirely a column-selection problem. Finally we contribute a string-structure fallback detector that covers columns the conditional model cannot represent, raising whole-table PR-AUC on one dataset from chance to 0.8360. We do not claim a competitive detector: against Raha at an equal 20-label budget we win one dataset of four.

Keywords*—*data cleaning, error detection, probability calibration, cost-sensitive decision making, abstention, class imbalance

I. INTRODUCTION

A dirty table contains cells whose values differ from the truth. The standard formulation asks a system to find those cells and correct them, and the standard metric rewards it for how often the correction matches the clean value. This framing hides a question that matters more in practice: when the system is unsure, what should it do?

The question is not rhetorical, because the two kinds of damage are not symmetric. Overwriting a correct value with a confident guess destroys information. Routing every suspicious cell to a human is safe but does not scale. Leaving an error in place is sometimes the cheapest option and sometimes the most expensive one, depending entirely on what the data is used for. A system that cannot express these trade-offs cannot be deployed responsibly, however accurate its repairs are.

We also found, repeatedly, that repair accuracy measures the dataset rather than the method. Corruption that mangles a value — an out-of-range number, a broken format — is recovered by simple rules 98–100% of the time. Corruption that erases a value is recovered 29–50% of the time by anything we tried, because the information is gone. Reporting a single repair accuracy therefore mostly reports which mixture of the two a benchmark happens to contain.

This paper makes three contributions, in descending order of how well the evidence supports them. First, a cost-sensitive decision layer that chooses among repairing, reviewing and abstaining using two calibrated probabilities, and which attains zero worst-case regret across the policies and datasets we tested. Second, a measured boundary: five mechanisms for column-level error localisation, each pre-registered with falsifying conditions, all refuted, with a common cause we identify. Third, a targeted fallback detector for high-cardinality text columns, which substantially improves coverage on some datasets and not others.

II. RELATED WORK

Rule-based and constraint-based cleaners such as NADEEF [1] and HoloClean [2] detect violations of user-supplied denial constraints and functional dependencies. Raha [3] runs many configurations of several base detectors, treats their outputs as a per-cell feature vector, and learns a classifier from roughly twenty labelled tuples; its companion Baran [4] handles repair. ED2 [5] adds active learning, including a component that selects which column to label next. HoloDetect [6] learns a corruption policy from the dirty table itself and trains a detector on synthesised errors, reporting that a learned augmentation policy substantially outperforms random transformations.

Two lines of work are closest to ours and bound our novelty claims. HoloDetect already calibrates detector scores using Platt scaling, so calibrating a detector is not itself new. Conformal Data Cleaning [7] pairs per-column conditional models with calibrated uncertainty, so per-column calibration is not new either. What we did not find in any of them is an evaluation of calibration quality — no reliability diagram, Brier score or expected calibration error appears — and none converts a calibrated score into a cost-weighted choice among three actions. That gap is where our contribution sits.

For the mixture formulation in Section V-B, the closest prior work is Kubica and Moore [8], who model a dirty table as a clean model plus a noise model with a corruption matrix fitted by alternating optimisation.

III. METHOD

A. Pipeline

The pipeline profiles each column, flags suspicious cells with six rule checks (missing tokens, out-of-range values, Tukey fences at k = 3, format violations, unseen categories, stray whitespace), proposes ranked repair candidates with probabilities, and derives three uncertainty signals: the top candidate's confidence, agreement across repeated sampling, and the normalised entropy of the candidate distribution.

B. Two calibrated probabilities

Raw confidence is not a probability. We fit a calibrator on a training split to obtain p, the probability that the top repair

candidate is correct, selecting among Platt scaling, isotonic regression and a multivariate variant by held-out Brier score.

A flag is also not proof. We fit a second calibrator, per detection check, to obtain q, the probability that a cell caught by that check is genuinely corrupted. This matters because a check can be systematically wrong: on one dataset every cell flagged by the Tukey fence is a statistical outlier and none is an error, so q approaches zero and the system correctly declines to repair clean data. Where a calibrator cannot be fitted, q defaults to 1, which reproduces the uncalibrated policy exactly and is a conservative assumption rather than evidence.

C. The decision rule

For costs C_wrong, C_human and C_abstain, the expected cost of each action is

$$E[repair] = (1 - p)\cdot C_wrong, \quad E[review] = C_human,$$
$$E[abstain] = q\cdot C_abstain$$

and the action taken is the arg min, with ties broken in the order repair, review, abstain. C_abstain is charged only when a genuinely corrupted cell is left unrepaired; any per-flag operational cost is decision-invariant and is reported separately rather than folded in. The costs are fixed constants read from a configuration file in abstract penalty units. They are not learned, not monetary, and not supplied by the benchmarks.

An honest limitation of the formulation: E[review] and E[abstain] are both constant in p, so for a single cell with fixed q one of the two is unreachable. What q changes is which of them is live, per cell, on evidence. All three actions being reachable is a claim about the population, not about every cell.

IV. EXPERIMENTAL SETUP

We evaluate on eleven datasets: three used throughout (Adult, Breast Cancer, Smart Factory) and eight standard error-detection benchmarks (Beers, Flights, Hospital, Movies, Rayyan, Restaurants, Soccer, Tax). Cell-level prevalence ranges from 0.00097 to 0.29582, which spans three orders of magnitude and is the regime that makes this problem hard.

Ground truth may be used only after predictions exist. The two calibrators consume 0/1 labels on a training split; all evaluation reads clean data after the fact. No detection, repair or decision function receives clean data, and automated tests assert this.

One further control deserves emphasis because it changed our conclusions. The column profile that supplies valid ranges, vocabularies and fences was originally built from clean data. We ablated this by rebuilding every profile statistic from the dirty table alone. The effect is reported in Section V-D and is large, so unless stated otherwise all detector numbers in this paper are the zero-knowledge ones.

Because five of our eleven datasets fall at or below the row cap used for subsampling, repeated row splits are not independent replicates on those datasets. We therefore vary the model seed across replicates and additionally report a 95% bootstrap confidence interval computed by resampling evaluated cells, which remains valid for a single run of a small table. The intervals in Table III are of that kind, and overlapping intervals are reported as overlapping rather than as improvements.

V. RESULTS

A. Cost-sensitive decisions

Table I reports realised cost in penalty units. Regret is the excess over the best policy for that dataset, so zero means the policy was never beaten.

TABLE I. REALISED COST AND WORST-CASE REGRET

Policy	Adult	Breast C.	Smart F.	Regret
Proposed	0	1,615	14,186	0
Always review	11,658	1,775	14,186	11,658
Confidence ≥ 0.70	11,658	1,720	14,186	11,658
Always abstain	0	3,550	35,465	21,279
Always repair	58,290	5,800	126,275	112,089

Every fixed rule is adequate on one dataset and badly beaten on another; the adaptive policy is the only one never badly beaten. It wins outright on Breast Cancer, ties always-abstain on Adult and ties always-review on Smart Factory; we report the ties as ties.

This claim is bounded by what we tested: three datasets, five candidate policies, one cost configuration per dataset, and this action space. It is not a claim of optimality.

Two results qualify it further. On Adult, once undetected corruption is priced, the proposed policy achieves total harm 8,505, which is exactly what doing nothing achieves: it avoids 11,658 to 58,290 units relative to every active baseline while adding nothing over having no detector at all. And the reported zero cost on Adult is residual-corruption cost on flagged cells only; its 1,701 genuine errors were never detected and enter every policy identically.

B. A measured boundary of detection

Our detector ranks cells well inside a column and poorly across columns. Scoring only the corrupted column of Adult gives PR-AUC 0.6873; scoring all fifteen gives 0.0123; zeroing the fourteen clean columns returns exactly 0.6873. The equality shows the loss is displacement, not dilution — rare but correct values in other columns outrank genuine errors.

We built five mechanisms to select the corrupted column, each pre-registered with conditions that could falsify it, and each was refuted. Table II records what each one measured instead.

TABLE II. REFUTED MECHANISMS AND THE QUANTITY EACH ABSORBED

Mechanism	Nuisance absorbed
Likelihood ratio	background density
Self-supervised injection	column cardinality
Contamination fraction	model competence
Contamination mixture	model error rate
Excess surprise	model miscalibration

The common cause is that every per-column statistic we constructed absorbed a column-specific nuisance quantity larger than the corruption signal. Each fix removed one nuisance and exposed the next. The mixture formulation is instructive: identifying corruption by its independence from the rest of the row is sound in principle, but on Adult it recovered a contamination rate of 0.1852 against a true rate of 0.0400, because 0.1852 is approximately the conditional

model's own error rate on that column. To an imperfect model, its own mistakes also look row-independent.

Because the corruption process is known for injected data, the Bayes-optimal posterior can be computed and gives an attainable ceiling. Within Adult's corrupted column that ceiling is PR-AUC 0.7222 and our detector reaches 0.5626, capturing 77.9%. Per-cell ranking inside a column is therefore close to exhausted, and essentially all remaining loss is column selection. Measuring what perfect column selection would be worth, however, shows it changes the whole-table result by 69.4× on Adult and only 1.0–1.7× on the other ten datasets, because those have corruption in most of their columns. Column selection is thus a far narrower problem than we assumed when we began.

C. Targeted text fallback

The conditional model treats a column as a categorical variable and skips high-cardinality text, which on real benchmarks is most of the table: it models 0 of 7 columns on Flights, 2 of 17 on Movies and 8 of 20 on Hospital. We add a detector that models the string rather than the category, combining shape rarity, character n-gram implausibility, and — the discriminating signal — similarity to the nearest frequent value. A typo sits close to a common value without equalling it, whereas a legitimately rare value sits far from everything. This inverts the usual rarity heuristic and is what separates the two cases.

TABLE III. WHOLE-TABLE PR-AUC, CONDITIONAL MODEL ALONE AND WITH FALLBACK

Dataset	Base	+ Fallback
Flights	0.2958 [.289,.302]	0.8360 [.828,.844]
Hospital	0.2632 [.225,.303]	0.4974 [.454,.542]
Beers	0.4104 [.395,.427]	0.5087 [.492,.526]
Rayyan	0.0931 [.086,.100]	0.1639 [.152,.177]
Tax	0.0739 [.014,.160]	0.0966 [.024,.192]
Movies	0.0872 [.081,.094]	0.0876 [.082,.094]

Flights is the clearest case: the conditional model scores no columns at all there, so its 0.2958 is simply the prevalence. On Flights and Hospital the intervals before and after are disjoint, so those gains are separated rather than merely larger point estimates. Elsewhere the picture is weaker and the scope must be stated plainly. On Tax and Movies the intervals overlap, so no improvement is claimed there; on two datasets a pure rarity baseline matches or beats the fallback; and the detector must be gated to abstain on columns where it cannot discriminate, because it otherwise scores columns containing no errors at all and displaces true positives in the ranking. This is a targeted extension for text-heavy tables, not a general detector.

D. Comparison with a published system, and the leakage ablation

Removing clean-derived profile statistics is severe for rule-based detection: Breast Cancer PR-AUC falls from 0.6712 to 0.5224 with non-overlapping intervals, and Hospital F1 falls from 0.9271 to 0.0000, because the novel-category and valid-range checks are by definition comparisons against a known-good vocabulary. Build that vocabulary from dirty data and the corrupted values are already inside it.

Against Raha at an equal 20-label budget, with failed label draws scored as failures rather than discarded, we win one dataset of four (Table IV). We therefore do not claim a competitive detector.

TABLE IV. F1 AT AN EQUAL 20-LABEL BUDGET

Dataset	Ours	Raha
Breast Cancer	0.3934	0.0404
Hospital	0.3679	0.4000
Beers	0.5001	0.7988
Rayyan	0.1079	0.6457

The Hospital loss is of a different kind from the other two. Our oracle-threshold F1 there is 0.5054, above Raha's 0.4000, so the detector is good enough and the budget is insufficient to locate the threshold. Allocating those twenty labels across score strata rather than uniformly raises F1 from 0.2014 to 0.3679. On Beers and Rayyan, Raha is better even than our oracle threshold, which is a genuine detection deficit.

VI. LIMITATIONS

We state these as findings rather than caveats. The zero-regret result is relative to three datasets, five policies, one cost configuration each and this action space. On Adult the system equals doing nothing once undetected errors are priced. The detector is not competitive with Raha under equal supervision. One result is unexplained: on Rayyan the combined system improves whole-table PR-AUC by 0.0708 although the single newly covered column scores below chance, and we do not yet know why. Five of our eleven datasets are too small for independent row splits, so uncertainty there rests on cell-level bootstrap rather than resampled runs. Finally, the costs are assumptions: we show that decisions change when they change, not that any particular setting is correct.

VII. CONCLUSION

Deciding whether to act is a better-posed problem than deciding what to write. A calibrated probability of correctness, a calibrated probability of corruption, and explicit costs give a policy that is never badly beaten on the datasets we tested, and that survives the removal of every clean-data advantage. The detection side of the same system is bounded in a way we can now describe precisely: inside a column it captures 77.9% of the attainable signal, and across columns it is defeated by nuisance quantities that five successive mechanisms each absorbed in turn. We think the measured boundary is as useful a contribution as the policy, and considerably more useful than another tuned detector.

. REFERENCES

[1] M. Dallachiesa et al., “NADEEF: A commodity data cleaning system,” in Proc. ACM SIGMOD, 2013, pp. 541–552.

[2] T. Rekatsinas, X. Chu, I. F. Ilyas, and C. Ré, “HoloClean: Holistic data repairs with probabilistic inference,” PVLDB, vol. 10, no. 11, pp. 1190–1201, 2017.

[3] M. Mahdavi et al., “Raha: A configuration-free error detection system,” in Proc. ACM SIGMOD, 2019, pp. 865–882.

[4] M. Mahdavi and Z. Abedjan, “Baran: Effective error correction via a unified context representation,” PVLDB, vol. 13, no. 12, pp. 1948–1961, 2020.

[5] F. Neutatz, M. Mahdavi, and Z. Abedjan, “ED2: A case for active learning in error detection,” in Proc. ACM CIKM, 2019, pp. 2249–2252.

[6] A. Heidari, J. McGrath, I. F. Ilyas, and T. Rekatsinas, “HoloDetect: Few-shot learning for error detection,” in Proc. ACM SIGMOD, 2019, pp. 829–846.

[7] S. Jäger and F. Bießmann, “From data imputation to data cleaning: Automated cleaning of tabular data improves downstream predictive performance,” in Proc. AISTATS, PMLR vol. 238, 2024.
[8] J. Kubica and A. Moore, “Probabilistic noise identification and data cleaning,” in Proc. IEEE ICDM, 2003, pp. 131–138.